Prompt-Powered Kickstart

Building a Beginner’s Toolkit for Flask

1. Title & Objective

Title: Getting Started with Flask – A Beginner’s Guide
Technology Chosen: Flask (Python Web Framework)
Reason for Choice:
I chose Flask because it’s lightweight, beginner-friendly and widely used to build web apps and APIs. It allows learning the basics of Python web development quickly.

End Goal:
Create a minimal “Hello World” API, learn to set up Flask projects and explore how AI can assist in coding and debugging.

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2. Overview of Flask

What is Flask?
Flask is a micro web framework for Python that allows developers to build web applications and APIs with minimal boilerplate code.

Use Cases:

Rapid prototyping

Learning web development

Backend APIs for small applications

Real-World Example:
Flask was used in early Instagram APIs and many small web services due to its simplicity and flexibility.

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3. Learning Process & AI Integration

Step 1: Scaffold Project with AI Help

Prompt Used: *“Give me a step-by-step guide to create a Hello World API with Flask”*

AI Response: Provided setup instructions, virtual environment commands and example code.

Reflection: The AI helped me start the project quickly without searching multiple tutorials.

Step 2: Running the API and Troubleshooting

Prompt Used: *"How to run Flask on a different port?"*

AI Response: Explained `app.run(port=5001)` to resolve port conflicts.

Reflection: Saved time fixing errors and helped me understand Flask's run method options.

Step 3: Understanding Code Flow

Explored Flask decorators (`@app.route('/')`) and functions.

AI suggested inline explanations and comments for beginners.

Reflection: Reinforced how routes work and how functions return HTTP responses.

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4. Minimal Working Example

File: `app.py`

```
from flask import Flask

app = Flask(__name__)

@app.route('/')
def hello():
    return "Hello World!"

if __name__ == "__main__":
    app.run(debug=True)
```

Explanation:

- `Flask(__name__)` initializes the app.
- `@app.route('/')` sets the URL endpoint.
- The `hello()` function returns the response text.
- `debug=True` enables live reload for development.

Reflection:
Building this minimal example taught me how simple it is to set up a web server in Python and test it locally.

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5. Common Issues & How I Resolved Them

Issue	Solution	Learning Reflection
ModuleNotFoundError: No module named 'flask'	`pip install Flask`	Understood Python package management and virtual environments.
Port 5000 already in use	`app.run(port=5001)`	Learned how to customize Flask server settings.
Virtual environment not activating	`venv\Scripts\activate`	Learned Windows-specific virtual environment commands.

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6. AI Prompt Journal

Prompt	AI Response Summary	Personal Reflection
"Step-by-step guide to create Hello World API with Flask"	Full setup instructions and sample code	Accelerated project initiation
"How to run Flask on a different port?"	Explained `app.run(port=5001)`	Solved a real-world error quickly
"Explain Flask decorators for beginners"	Provided clear explanation with examples	Reinforced understanding of route handling

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7. References & Learning Resources

- Flask Official Documentation
- RealPython Flask Tutorial
- StackOverflow Flask Questions
- AI prompts from ai.moringaschool.com
